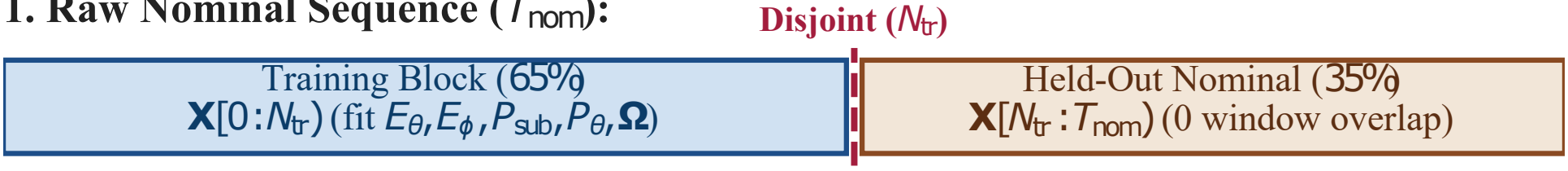
1. Raw Nominal Sequence (T_{nom}):



2. Held-Out Causal Indexing ($W = C + S = 320$, $\tau = t + S - 1$):



At gap $S=64$: suspect targets $\mathbf{x}_{tgt}$ are 100% disjoint ($W=320$ check: $\Delta\text{FPR} \le 0.002$).

3. Online Window Causality ($\vec{\tau}$): Emission Lag vs. Detection Delay:

